METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

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HEAT AND MATERIAL BALANCE

R-201 quench converter and synthesis loop · design and three off-design cases

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DOCUMENT TITLE	:	Heat and Material Balance
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Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

1 Purpose and scope

This document gives the heat and material balance of the Methanol Converter Ver 1 synthesis loop (battery limits: make-up syngas at the syngas-compressor discharge, purge gas to the reformer fuel header, crude methanol to let-down) for the design case and three off-design cases. Stream numbers and names are those of PFD-201 and the Process tab (streams 1–12). Every number is computed by the simulator model (tabs: Process, Design of the unit operation, Operating windows); nothing is taken from a vendor or a real plant.

The off-design cases are steady states reached by the model's owRun routine (a copy of the model, noise off, trips disabled so that a limit is seen rather than cut short, 6000 s simulated); the design case is the converged design snapshot used by every tab. The governing values for line sizing (MEOH-V1-PE-LSL-001) and the design margins (section 8) are taken over all four cases.

2 Cases

Case	What it is	Make-up kNm³/h	Circulation kNm³/h	Purge kNm³/h	V-201 bar g	Conv. inlet bar g	CH3OH t/h	Hottest catalyst °C	C-201 speed % / kW	Limits crossed (owEval)
Design (normal)	Design snapshot of the simulator (warm start 1200 s, noise off) = the page's snapDesign: make-up 365.6 kNm³/h, circulation 1928 kNm³/h, V-201 75 bar g, bed inlets 220/225/230/235 °C, fresh catalyst.	365.6	1,928	105.1	75.00	79.70	112.5	276.9	80.0 / 6,528	none
Turndown 70 %	Make-up FIC-202 at 70 % (256 kNm³/h), circulation FIC-201 at 80 % (1542 kNm³/h, recycle ratio kept near design), other set points unchanged.	256.0	1,542	69.0	75.00	78.01	80.7	270.8	64.0 / 3,348	none
End of run (low activity)	Catalyst ageing at full severity (every bed at 40 % of fresh activity), bed inlet set points raised 10 °C to 230/235/240/245 °C as an operator would at end of run; make-up and circulation at design.	365.6	1,928	125.5	75.00	79.70	103.8	280.3	80.0 / 6,504	none
Maximum rate 110 %	Make-up FIC-202 at 110 % (402 kNm³/h) and circulation FIC-201 at 109 % (2100 kNm³/h); fresh catalyst, design bed inlets. Used as the rated case for lines and equipment.	402.0	2,100	117.2	75.00	80.58	123.0	277.3	87.1 / 8,414	none

Limits crossed: the bound list of the Operating windows tab (trip margins, valve ranges, alarms) evaluated on the steady state; "none" means every bound is respected. All four cases hold PIC-203 at 75 bar g and the bed inlets on their set points.

Make-up gas (battery limit, assumed in the model)

H2 73 · CO 16 · CO2 7 · CH4 3.3 · N2 0.5 · H2O 0.2 mol %, 40 °C, 86 bar a upstream of FV-202. Converter catalyst 184 m³ Cu/ZnO/Al2O3 split 12/20/28/40 %, kinetics of Vanden Bussche & Froment (1996) (model assumptions, see the Workbook).

3.1 Stream table · Design (normal)

Design snapshot of the simulator (warm start 1200 s, noise off) = the page's snapDesign: make-up 365.6 kNm³/h, circulation 1928 kNm³/h, V-201 75 bar g, bed inlets 220/225/230/235 °C, fresh catalyst.

Stream	Unit	1	2	3	4	5	6	7	8	9	10	11	12
Name		Make-up syngas	Recycle gas to C-201	Converter feed (C-201 discharge)	Feed after E-202 (= quench gas)	Bed 1 feed (after E-201, E-204)	Cold shots to beds 2-4	Converter effluent	Effluent after E-201	Effluent after E-202 (to E-203)	V-201 vapour	Purge gas to fuel	Crude methanol (liquid)
Phase		vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	liquid
Molar flow	kmol/h	16,311	69,706	86,018	86,018	40,757	45,261	78,962	78,962	78,962	74,395	4,689	4,567
Gas flow (0 °C, 1 atm)	kNm³/h	365.6	1,562.4	1,928.0	1,928.0	913.5	1,014.5	1,769.9	1,769.9	1,769.9	1,667.5	105.1	—
Mass flow	kg/h	158,861	406,411	565,272	565,272	267,836	297,436	565,264	565,264	565,264	433,748	27,337	131,516
Temperature	°C	40.0	40.0	48.6	96.8	220.0	96.8	267.6	206.4	155.9	40.0	40.0	40.0
Pressure	bar g	85.99	75.00	80.60	79.70	79.70	79.70	77.37	76.97	76.67	75.00	75.00	75.00
Mol. weight	kg/kmol	9.74	5.83	6.57	6.57	6.57	6.57	7.16	7.16	7.16	5.83	5.83	28.80
Density	kg/m³	32.54	17.02	20.05	17.25	12.94	17.25	12.48	14.00	15.59	17.02	17.02	814.30
Actual volume flow	m³/h	4,882	23,876	28,194	32,777	20,705	17,247	45,295	40,374	36,262	25,482	1,606	162
Cp (model mean)	kJ/kmol K	31.2	31.8	31.7	31.7	31.7	31.7	32.9	32.9	32.9	31.8	31.8	—
Viscosity (assumed)	cP	0.0130	0.0130	0.0132	0.0146	0.0179	0.0146	0.0191	0.0175	0.0162	0.0130	0.0130	0.5500
Composition	mol %												
H2	mol %	73.00	81.62	79.99	79.99	79.99	79.99	76.90	76.90	76.90	81.62	81.62	0.000
CO	mol %	16.00	2.27	4.88	4.88	4.88	4.88	2.14	2.14	2.14	2.27	2.27	0.000
CO2	mol %	7.00	2.50	3.35	3.35	3.35	3.35	2.35	2.35	2.35	2.50	2.50	0.000
CH3OH	mol %	0.000	0.358	0.290	0.290	0.290	0.290	4.78	4.78	4.78	0.358	0.358	76.88
H2O	mol %	0.200	0.022	0.056	0.056	0.056	0.056	1.36	1.36	1.36	0.022	0.022	23.12
CH4	mol %	3.30	11.48	9.93	9.93	9.93	9.93	10.82	10.82	10.82	11.48	11.48	0.000
N2	mol %	0.500	1.74	1.50	1.50	1.50	1.50	1.64	1.64	1.64	1.74	1.74	0.000
CH3OH in liquid	wt %												85.5

3.2 Stream table · Turndown 70 %

Make-up FIC-202 at 70 % (256 kNm³/h), circulation FIC-201 at 80 % (1542 kNm³/h, recycle ratio kept near design), other set points unchanged.

Stream	Unit	1	2	3	4	5	6	7	8	9	10	11	12
Name		Make-up syngas	Recycle gas to C-201	Converter feed (C-201 discharge)	Feed after E-202 (= quench gas)	Bed 1 feed (after E-201, E-204)	Cold shots to beds 2-4	Converter effluent	Effluent after E-201	Effluent after E-202 (to E-203)	V-201 vapour	Purge gas to fuel	Crude methanol (liquid)
Phase		vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	liquid
Molar flow	kmol/h	11,421	57,375	68,796	68,796	37,418	31,378	63,740	63,740	63,740	60,452	3,077	3,289
Gas flow (0 °C, 1 atm)	kNm³/h	256.0	1,286.0	1,542.0	1,542.0	838.7	703.3	1,428.7	1,428.7	1,428.7	1,355.0	69.0	—
Mass flow	kg/h	111,238	310,871	422,108	422,108	229,586	192,523	422,102	422,102	422,102	327,541	16,671	94,561
Temperature	°C	40.0	37.7	43.6	90.3	220.0	90.3	261.2	187.6	138.9	37.7	37.7	37.7
Pressure	bar g	85.99	75.00	78.58	78.01	78.01	78.01	76.33	75.93	75.63	75.00	75.00	75.00
Mol. weight	kg/kmol	9.74	5.42	6.14	6.14	6.14	6.14	6.62	6.62	6.62	5.42	5.42	28.75
Density	kg/m³	32.54	15.94	18.54	16.04	11.82	16.04	11.53	13.30	14.82	15.94	15.94	814.60
Actual volume flow	m³/h	3,418	19,508	22,762	26,312	19,416	12,001	36,612	31,735	28,488	20,555	1,046	116
Cp (model mean)	kJ/kmol K	31.2	31.8	31.7	31.7	31.7	31.7	32.8	32.8	32.8	31.8	31.8	—
Viscosity (assumed)	cP	0.0130	0.0129	0.0131	0.0144	0.0179	0.0144	0.0189	0.0170	0.0158	0.0129	0.0129	0.5500
Composition	mol %												
H2	mol %	73.00	82.34	80.79	80.79	80.79	80.79	78.09	78.09	78.09	82.34	82.34	0.000
CO	mol %	16.00	1.54	3.94	3.94	3.94	3.94	1.47	1.47	1.47	1.54	1.54	0.000
CO2	mol %	7.00	1.67	2.55	2.55	2.55	2.55	1.58	1.58	1.58	1.67	1.67	0.000
CH3OH	mol %	0.000	0.320	0.267	0.267	0.267	0.267	4.25	4.25	4.25	0.320	0.320	76.57
H2O	mol %	0.200	0.020	0.050	0.050	0.050	0.050	1.23	1.23	1.23	0.020	0.020	23.43
CH4	mol %	3.30	12.25	10.76	10.76	10.76	10.76	11.62	11.62	11.62	12.25	12.25	0.000
N2	mol %	0.500	1.86	1.63	1.63	1.63	1.63	1.76	1.76	1.76	1.86	1.86	0.000
CH3OH in liquid	wt %												85.3

3.3 Stream table · End of run (low activity)

Catalyst ageing at full severity (every bed at 40 % of fresh activity), bed inlet set points raised 10 °C to 230/235/240/245 °C as an operator would at end of run; make-up and circulation at design.

Stream	Unit	1	2	3	4	5	6	7	8	9	10	11	12
Name		Make-up syngas	Recycle gas to C-201	Converter feed (C-201 discharge)	Feed after E-202 (= quench gas)	Bed 1 feed (after E-201, E-204)	Cold shots to beds 2-4	Converter effluent	Effluent after E-201	Effluent after E-202 (to E-203)	V-201 vapour	Purge gas to fuel	Crude methanol (liquid)
Phase		vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	liquid
Molar flow	kmol/h	16,311	69,706	86,018	86,018	51,982	34,035	79,499	79,499	79,499	75,306	5,599	4,193
Gas flow (0 °C, 1 atm)	kNm³/h	365.6	1,562.4	1,928.0	1,928.0	1,165.1	762.9	1,781.9	1,781.9	1,781.9	1,687.9	125.5	—
Mass flow	kg/h	158,861	471,501	630,362	630,362	380,941	249,421	630,355	630,355	630,355	509,376	37,875	120,979
Temperature	°C	40.0	39.4	48.1	92.5	230.0	92.5	280.3	193.5	147.1	39.4	39.4	39.4
Pressure	bar g	85.99	75.00	80.60	79.70	79.70	79.70	76.97	76.57	76.27	75.00	75.00	75.00
Mol. weight	kg/kmol	9.74	6.76	7.33	7.33	7.33	7.33	7.93	7.93	7.93	6.76	6.76	28.85
Density	kg/m³	32.54	19.78	22.39	19.45	14.14	19.45	13.44	15.85	17.54	19.78	19.78	813.80
Actual volume flow	m³/h	4,882	23,833	28,152	32,402	26,943	12,821	46,910	39,759	35,945	25,747	1,914	149
Cp (model mean)	kJ/kmol K	31.2	31.8	31.6	31.6	31.6	31.6	32.8	32.8	32.8	31.8	31.8	—
Viscosity (assumed)	cP	0.0130	0.0130	0.0132	0.0145	0.0181	0.0145	0.0194	0.0172	0.0160	0.0130	0.0130	0.5500
Composition	mol %												
H2	mol %	73.00	79.77	78.48	78.48	78.48	78.48	75.56	75.56	75.56	79.77	79.77	0.000
CO	mol %	16.00	4.87	6.98	6.98	6.98	6.98	4.62	4.62	4.62	4.87	4.87	0.000
CO2	mol %	7.00	3.92	4.50	4.50	4.50	4.50	3.71	3.71	3.71	3.92	3.92	0.000
CH3OH	mol %	0.000	0.351	0.284	0.284	0.284	0.284	4.41	4.41	4.41	0.351	0.351	77.25
H2O	mol %	0.200	0.021	0.055	0.055	0.055	0.055	1.22	1.22	1.22	0.021	0.021	22.75
CH4	mol %	3.30	9.61	8.42	8.42	8.42	8.42	9.11	9.11	9.11	9.61	9.61	0.000
N2	mol %	0.500	1.46	1.27	1.27	1.27	1.27	1.38	1.38	1.38	1.46	1.46	0.000
CH3OH in liquid	wt %												85.8

3.4 Stream table · Maximum rate 110 %

Make-up FIC-202 at 110 % (402 kNm³/h) and circulation FIC-201 at 109 % (2100 kNm³/h); fresh catalyst, design bed inlets. Used as the rated case for lines and equipment.

Stream	Unit	1	2	3	4	5	6	7	8	9	10	11	12
Name		Make-up syngas	Recycle gas to C-201	Converter feed (C-201 discharge)	Feed after E-202 (= quench gas)	Bed 1 feed (after E-201, E-204)	Cold shots to beds 2-4	Converter effluent	Effluent after E-201	Effluent after E-202 (to E-203)	V-201 vapour	Purge gas to fuel	Crude methanol (liquid)
Phase		vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	vapour	liquid
Molar flow	kmol/h	17,935	75,756	93,691	93,691	44,498	49,193	85,975	85,975	85,975	80,984	5,228	4,990
Gas flow (0 °C, 1 atm)	kNm³/h	402.0	1,698.0	2,100.0	2,100.0	997.4	1,102.6	1,927.0	1,927.0	1,927.0	1,815.2	117.2	—
Mass flow	kg/h	174,678	448,117	622,795	622,795	295,792	327,003	622,786	622,786	622,786	479,043	30,926	143,743
Temperature	°C	40.0	40.8	50.8	97.6	220.0	97.6	268.8	207.8	158.8	40.8	40.8	40.8
Pressure	bar g	85.99	75.00	81.64	80.58	80.58	80.58	77.83	77.43	77.13	75.00	75.00	75.00
Mol. weight	kg/kmol	9.74	5.92	6.65	6.65	6.65	6.65	7.24	7.24	7.24	5.92	5.92	28.80
Density	kg/m³	32.54	17.23	20.40	17.59	13.23	17.59	12.67	14.21	15.76	17.23	17.23	814.20
Actual volume flow	m³/h	5,368	26,012	30,529	35,398	22,363	18,586	49,135	43,831	39,509	27,808	1,795	176
Cp (model mean)	kJ/kmol K	31.2	31.8	31.7	31.7	31.7	31.7	33.0	33.0	33.0	31.8	31.8	—
Viscosity (assumed)	cP	0.0130	0.0130	0.0133	0.0146	0.0179	0.0146	0.0191	0.0176	0.0163	0.0130	0.0130	0.5500
Composition	mol %												
H2	mol %	73.00	81.47	79.84	79.84	79.84	79.84	76.74	76.74	76.74	81.47	81.47	0.000
CO	mol %	16.00	2.45	5.05	5.05	5.05	5.05	2.31	2.31	2.31	2.45	2.45	0.000
CO2	mol %	7.00	2.65	3.48	3.48	3.48	3.48	2.50	2.50	2.50	2.65	2.65	0.000
CH3OH	mol %	0.000	0.372	0.300	0.300	0.300	0.300	4.82	4.82	4.82	0.372	0.372	76.92
H2O	mol %	0.200	0.023	0.057	0.057	0.057	0.057	1.36	1.36	1.36	0.023	0.023	23.08
CH4	mol %	3.30	11.32	9.79	9.79	9.79	9.79	10.66	10.66	10.66	11.32	11.32	0.000
N2	mol %	0.500	1.72	1.48	1.48	1.48	1.48	1.62	1.62	1.62	1.72	1.72	0.000
CH3OH in liquid	wt %												85.6

4 Converter R-201 bed by bed

Case	Bed	Gas in, kNm³/h	Inlet bar g	T in °C	T out °C	Rise °C	Hottest cell °C	CH3OH in %	CH3OH out %	ΔP bar
Design	1	914	79.70	220.0	260.0	40.0	260.0	0.29	1.96	0.307
	2	1,158	79.39	225.0	276.9	51.9	276.9	1.60	3.59	0.477
	3	1,551	78.92	230.0	270.0	40.0	270.0	2.71	4.29	0.659
	4	1,928	78.26	235.0	267.6	32.6	267.6	3.46	4.78	0.885
Turndown	1	839	78.01	220.0	252.7	32.7	252.7	0.27	1.64	0.261
	2	1,008	77.75	225.0	270.8	45.8	270.8	1.41	3.14	0.368
	3	1,294	77.38	230.0	263.8	33.8	263.8	2.48	3.81	0.466
	4	1,542	76.91	235.0	261.2	26.2	261.2	3.20	4.25	0.577
End of run	1	1,165	79.70	230.0	250.3	20.3	250.3	0.28	1.24	0.513
	2	1,289	79.19	235.0	275.6	40.6	275.6	1.14	2.65	0.613
	3	1,592	78.57	240.0	278.4	38.4	278.4	2.18	3.64	0.711
	4	1,928	77.86	245.0	280.3	35.3	280.3	3.03	4.41	0.897
Max rate	1	997	80.58	220.0	257.8	37.8	257.8	0.30	1.89	0.367
	2	1,249	80.21	225.0	277.3	52.3	277.3	1.56	3.57	0.556
	3	1,680	79.65	230.0	270.9	40.9	270.9	2.69	4.30	0.773
	4	2,100	78.88	235.0	268.8	33.8	268.8	3.45	4.82	1.049

Gas in = bed-1 feed plus the cold shots added above the bed. CH3OH in the gas, mol %. ΔP per bed from the model (scales with flow² and 1/pressure).

5 Unit duties

Unit	Duty	Unit	Design	Turndown	End of run	Max rate
R-201	Heat released by the reactions (XI-260)	MW	84.65	60.51	78.48	92.62
E-201	Hot feed/effluent interchanger	MW	44.23	42.72	62.81	47.96
E-202	Cold feed/effluent interchanger	MW	36.49	28.32	33.58	38.62
E-203	Final cooler, sensible (the model's E-203 heat)	MW	83.76	58.75	77.94	92.84
E-203	Final cooler, condensing crude methanol (latent heats assumed)	MW	49.18	35.43	45.13	53.73
E-203	Final cooler, total	MW	132.94	94.17	123.07	146.57
E-203	Cooling-water return TI-254 (model, sensible only)	°C	41.31	38.53	40.66	42.32
E-204	Start-up heater (out of service)	MW	0.00	0.00	0.00	0.00
C-201	Circulator shaft power (JI-201)	kW	6,528	3,348	6,504	8,414
C-201	Speed (SI-201)	%	80.0	64.0	80.0	87.1

The cooling-water flow implied by the model is constant (about 7 740 t/h for a 9 000 kW/K water heat-capacity rate). With the condensing duty included the return would be about 47 °C at design; see EQ-ep-07.

6 Balances

6.1 Overall mass balance

Case	Make-up in, t/h	Purge out, t/h	Crude methanol out, t/h	In − out, t/h	Closure, % of input	Model imbalance heat / carbon / mass, %
Design (normal)	158.86	27.34	131.52	0.008	0.005	0.0003 / 0.0001 / 0.0050
Turndown 70 %	111.24	16.67	94.56	0.006	0.005	0.0001 / 0.0000 / 0.0052
End of run (low activity)	158.86	37.88	120.98	0.007	0.005	0.0000 / 0.0000 / 0.0047
Maximum rate 110 %	174.68	30.93	143.74	0.009	0.005	0.0000 / 0.0000 / 0.0051

Model imbalance: the accumulation terms of the Process tab balances (balancesOf), as % of the input; small values show the case is at steady state.

6.2 Component balance · Design (normal)

Component	In (make-up), kmol/h	Generated (reactions), kmol/h	Out in purge, kmol/h	Out in crude, kmol/h	In + gen − out, kmol/h	Residual, %
CO	2,609.8	−2,503.1	106.7	0.0	0.00	0.000
CO2	1,141.8	−1,024.6	117.2	0.0	0.00	0.000
H2	11,907.2	−8,080.0	3,827.2	0.0	−0.02	0.000
CH3OH	0.0	3,527.7	16.8	3,510.9	0.00	0.000
H2O	32.6	1,024.6	1.1	1,056.2	0.00	0.000
CH4	538.3	0.0	538.3	0.0	0.00	−0.001
N2	81.6	0.0	81.6	0.0	0.00	−0.001

Generated = converter outlet − converter feed (the two reactions CO2 + 3H2 ⇌ CH3OH + H2O and CO2 + H2 ⇌ CO + H2O). The model's flash puts no dissolved gas in the crude methanol.

6.3 Element balance (in = make-up; out = purge + crude methanol)

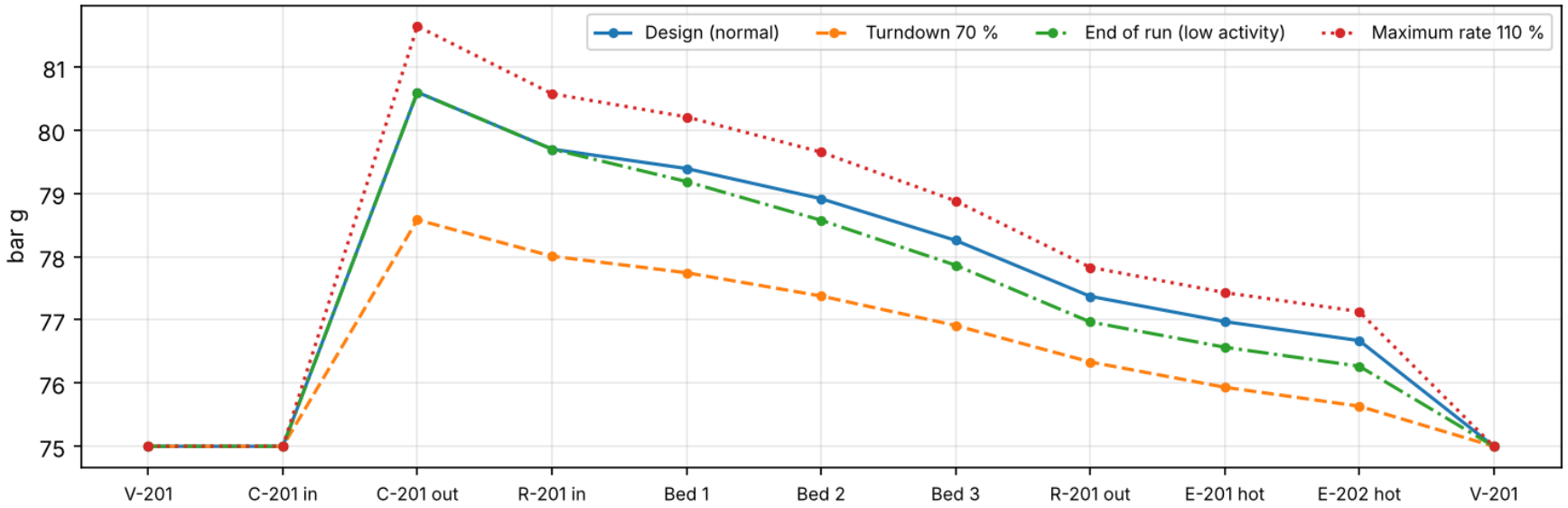
Element	Unit	Design in	Turndown in	End of run in	Max rate in	Design closure %	Turndown closure %	End of run closure %	Max rate closure %
C	kmol/h	4,289.9	3,003.8	4,289.9	4,717.0	0.000	0.000	0.000	0.000
H	kmol/h	26,032.7	18,228.6	26,032.7	28,624.6	0.000	0.000	0.000	0.000
O	kmol/h	4,926.0	3,449.3	4,926.0	5,416.4	0.000	0.000	0.000	0.000
N	kmol/h	163.1	114.2	163.1	179.4	−0.001	0.000	0.000	0.000

7 C-201 and loop hydraulics

7.1 Pressure profile (the model's own pressure drops)

Point round the loop	Unit	Design	Turndown	End of run	Max rate
V-201 separator (PIC-203)	bar g	75.00	75.00	75.00	75.00
C-201 suction (suction line ΔP not modelled)	bar g	75.00	75.00	75.00	75.00
C-201 discharge (PI on stream 3)	bar g	80.60	78.58	80.60	81.64
After E-202 cold side + lines = converter inlet PI-201	bar g	79.70	78.01	79.70	80.58
After bed 1	bar g	79.39	77.75	79.19	80.21
After bed 2	bar g	78.92	77.38	78.58	79.65
After bed 3	bar g	78.26	76.91	77.86	78.88
Converter outlet (after bed 4)	bar g	77.37	76.33	76.97	77.83
After E-201 hot side (stream 8)	bar g	76.97	75.93	76.57	77.43
After E-202 hot side (stream 9)	bar g	76.67	75.63	76.27	77.13
V-201 inlet (E-203 + lines by difference)	bar g	75.00	75.00	75.00	75.00
Loop ΔP = C-201 differential	bar	5.60	3.58	5.60	6.64
of which R-201 beds	bar	2.33	1.67	2.73	2.75
of which E-202 cold side + lines	bar	0.90	0.58	0.90	1.07
of which E-203 + lines to V-201 (by difference)	bar	1.67	0.63	1.27	2.13

Model: loop ΔP = 5.6 bar × (F/F_des)² × (P_ref/P_sep); E-202 cold side and the converter inlet lines 0.9 bar of it; bed ΔP from the bed model; E-201 hot side 0.4 bar and E-202 hot side 0.3 bar are the Process-tab stream-table conventions; the rest (E-203, V-201 inlet, suction) is by difference.



7.2 C-201 head and power at design

Step	Value	How
Suction pressure P1	75.00 bar g = 76.01 bar a	V-201 pressure (suction line loss not modelled)
Discharge pressure P2	80.60 bar g	P1 + loop ΔP 5.6 bar
Pressure ratio r = P2/P1 (absolute)	1.0737	
Suction temperature T1	40.0 °C	recycle at V-201 temperature, make-up at 40 °C
Model temperature rise ΔT = T1 (r^0.275 − 1) / 0.72	8.59 K	model: exponent 0.275, isentropic efficiency 0.72 (built into the model)
Model discharge temperature TI-202	48.58 °C	T1 + ΔT (stream 3)
Model power = F · cp(loop gas) · ΔT	6,528 kW (model JI-201: 6,528 kW)	F = 23.894 kmol/s, cp = 31.82 kJ/(kmol K)
Hand check: suction gas MW, cp, k	6.572 kg/kmol, 31.70 kJ/(kmol K), k = 1.356	converter-feed composition (stream 3); k = cp/(cp − R)
Isentropic head H = cp/MW · T1 · (r^((k−1)/k) − 1)	28.43 kJ/kg = 2,899 m	ideal gas, Z = 1
Mass flow	157.0 kg/s = 565 t/h	stream 3
Gas power = m · H / 0.72	6,200 kW	−5.0 % against the model
Actual suction volume	29,464 m³/h	F · R · T1 / P1

The hand check differs from the model by a few per cent because the model uses a fixed exponent 0.275 (k ≈ 1.38) where the mixture cp gives k ≈ 1.36; both are within the accuracy of a conceptual circulator rating. Motor rating and the electrical load are in MEOH-V1-EE-LL-001.

8 Design margins

Normal = the design case of this HMB (the model's design point). Max case = the highest value over the four cases. Design = the larger of normal + typical margin and the max case. Typical margins are assumed (10 % on flow, power and duty; 20 % on the purge, which grows as the catalyst ages). Capacity = what the model can deliver with the valve fully open or the machine at 100 % speed, and its margin over the design value.

Item	Unit	Normal	Max case	Governing case	Typical margin (assumed)	Design	Design basis	Capacity in the model	Capacity over design	Capacity basis
Make-up gas FIC-202 / FV-202	kNm³/h	365.6	402.0	Max rate	10 %	402.2	normal + 10 %	731.3	81.8 %	FV-202 fully open at 75 bar g (model K.Km)
Circulation C-201 (FIC-201)	kNm³/h	1,928.0	2,100.0	Max rate	10 %	2,120.8	normal + 10 %	2,410.2	13.6 %	C-201 at 100 % speed (model K.F100)
C-201 shaft power	kW	6,528.0	8,414.0	Max rate	10 %	8,414.0	max case (Max rate)	12,576.6	49.5 %	model formula at 100 % speed
Heat of reaction R-201	MW	84.7	92.6	Max rate	10 %	93.1	normal + 10 %	—	—	—
E-201 hot interchanger duty	MW	44.2	62.8	End of run	10 %	62.8	max case (End of run)	—	—	UA fixed in the model (K.UA201)
E-202 cold interchanger duty	MW	36.5	38.6	Max rate	10 %	40.1	normal + 10 %	—	—	UA fixed in the model (K.UA202)
E-203 final cooler duty (sensible + latent)	MW	132.9	146.6	Max rate	10 %	146.6	max case (Max rate)	—	—	latent heat assumed
Purge PV-203	kNm³/h	105.1	125.5	End of run	20 %	126.1	normal + 20 %	233.6	85.2 %	PV-203 fully open at 75 bar g (model K.Kpu)
Crude methanol LV-204	m³/h	161.5	176.5	Max rate	10 %	177.7	normal + 10 %	323.0	81.8 %	LV-204 fully open, 75 → 6 bar g (model K.Klv)
Methanol production	t/h	112.5	123.0	Max rate	0 %	123.0	max case (Max rate)	—	—	plant rate 112.5 t/h (900 kt/y)

8.1 Design pressure and temperature basis

Item	Max operating (HMB)	Unit	HMB rule (typical, assumed)	HMB minimum	Design (MEOH-V1-ME-EQL-001)	EQL item	EQL ≥ HMB	Trip / alarm
Zone A: V-201, C-201 suction, purge and crude lines	75.0	bar g	max(op + 10 %, op + 1.7 bar)	82.5	88	zone A = PSV-201 set	yes	PSHH-203 85 bar g
Zone B: C-201 discharge → E-202 → R-201 → E-201 → E-203 → V-201 inlet	81.6	bar g	PSV-201 set + loop ΔP with C-201 running (before accumulation)	92.8	100	zone B (110 % of set + C-201 ΔP at 100 % speed)	yes	—
R-201 (hottest catalyst)	280.3	°C	TSHH trip + 25 °C	325.0	335	R-201	yes	TSHH-2xx 300 °C (I-1)
Converter effluent DN900-PG-251 / E-201 tube inlet	280.3	°C	max op + 25 °C, not below the TSHH	306.0	335	E-201 tube side	yes	—
E-202 tube inlet (stream 8)	207.8	°C	max op + 25 °C	233.0	280	E-202 tube side	yes	—
E-203 inlet (stream 9)	158.8	°C	max op + 25 °C	184.0	225	E-203 tube side	yes	—
Converter feed after E-202 / quench (stream 4)	97.6	°C	max op + 25 °C	123.0	170	E-202 shell side	yes	—
C-201 discharge (stream 3)	50.8	°C	max op + 25 °C	76.0	120	C-201 casing	yes	—
V-201 / C-201 suction	40.8	°C	max op + 25 °C	66.0	120	V-201	yes	TI-253 high alarm 45 °C

The design pressures and temperatures are those of the equipment list MEOH-V1-ME-EQL-001 (zone A 88 bar g = PSV-201 set; zone B 100 bar g); the HMB column is the minimum this balance would set by the typical rule. The line sizing list MEOH-V1-PE-LSL-001 takes each line's design P/T from the same zones.

9 Engineering queries raised by this document

No.	To	Issue	Proposed	Status
EQ-ep-03	MEOH-V1-PE-RLB-001 / MEOH-V1-ME-EQL-001	PSV-201 (set 88 bar g) is on V-201, the low-pressure end of the loop. With C-201 running at design circulation and V-201 at the PSV set, the C-201 discharge side (E-202 cold side, R-201, E-201) sits at about 92.8 bar g, above the 88 bar g that the V-201 design pressure is based on.	Set the design pressure of the C-201 discharge side at PSV set + loop ΔP (≈ 93 bar g, before accumulation) or add relief on the C-201 discharge; confirm in the relief load basis and the equipment list.	answered by MEOH-V1-ME-EQL-001 (zone B 100 bar g)
EQ-ep-04	MEOH-PID-001 / -002 (design-point table)	The P&ID design-point table gives outlet CH3OH 2.1 / 3.6 / 4.4 / 4.8 % for beds 1–4; the model at design gives 1.96 / 3.59 / 4.29 / 4.78 %, i.e. 2.0 / 3.6 / 4.3 / 4.8 % to the table's one decimal: bed 1 and bed 3 differ. Bed temperatures agree within 0.5 °C.	Revise the P&ID design-point table CH3OH column to 2.0 / 3.6 / 4.3 / 4.8 % at the next revision (the HMB governs).	open
EQ-ep-05	MEOH-PID-002 note 1	Note 1 labels H2 81.6 / CH4 11.5 / CO2 2.5 / CO 2.3 / N2 1.7 / CH3OH 0.36 mol % as the loop gas "at C-201 suction"; that is the V-201 vapour (stream 10: H2 81.6, CH4 11.5). The make-up joins upstream of C-201, so the suction gas is stream 3: H2 80.0 / CH4 9.9 / CO2 3.4 / CO 4.9 / N2 1.5 / CH3OH 0.29 mol %.	Relabel note 1 as "V-201 vapour / recycle gas" or quote the stream 3 composition for C-201 suction.	open
EQ-ep-07	PFD tab equipment list / OTS model (E-203)	The PFD tab lists E-203 as "≈ 84 MW sensible + condensing". The HMB gives 83.76 MW sensible (the model's E-203 heat, which also sets the cooling-water return TI-254) plus 49.18 MW to condense the crude methanol (latent heats assumed), 132.94 MW in all. The model leaves the latent part out, so TI-254 reads 41.31 °C where the full duty would give about 46.8 °C at the same cooling-water flow.	Quote E-203 as ≈ 84 MW sensible + ≈ 49 MW condensing (≈ 133 MW) on the PFD tab; size E-203 and the cooling water on the full duty; record the sensible-only TI-254 as a known model simplification.	open

10 Assumptions and notes

1. Gas density from the ideal-gas law (Z = 1), as the model computes the loop pressure; a real H2-rich gas at 80 bar has Z ≈ 1.03–1.05, which lowers ρ and raises v by the same few per cent.

2. Gas viscosity 0.0130 cP at 40 °C × (T/313 K)^0.7; crude methanol 0.55 cP at 40 °C (assumed; not in the model).

3. Liquid densities 790 kg/m³ methanol and 995 kg/m³ water, volume-additive (the model's V-201 level calculation).

4. Latent heats 37.4 kJ/mol methanol and 43.3 kJ/mol water at 40 °C for the E-203 condensing duty (assumed; the model's E-203 heat is sensible only).

5. Line design pressure and temperature from MEOH-V1-ME-EQL-001: zone A 88 bar g (PSV-201 set), zone B 100 bar g; design temperature of the equipment each line connects. Pipe material carbon steel A106 Gr B (S = 138 MPa ≤ 204 °C, 130 MPa to 260 °C) up to the EQL carbon-steel limit in hot hydrogen (220 °C), 1¼Cr-½Mo A335 P11 (S = 125 MPa) above it; flanges Class 600 zone A, Class 900 zone B (all assumed).

6. Pipe walls: ASME B36.10 Sch 80 where listed and sufficient, else the ASME B31.3 304.1.2 minimum at the line design pressure, E = 1, Y = 0.4, + 1.5 mm corrosion allowance, ÷ 0.875 mill tolerance, rounded up to 1 mm (assumed; piping class to confirm).

7. Roughness 0.046 mm (commercial steel), Darcy friction factor from the Churchill (1977) equation.

8. Rated (design) flow for lines = the larger of normal × 1.10 and the highest flow of the four cases (typical, assumed).

9. Reformer fuel-header pressure 4 bar g (assumed, not in the model).

10. Margins: 10 % on flow, power and duty, 20 % on the purge (typical practice, assumed).

11. Turndown: the FSLL-201 trip is at 45 % of design circulation (868 kNm³/h); the turndown case runs at 1,542 kNm³/h, 80 % of design.

12. The data come from epc/eproc_data.js (node, model and tabs/shared/ow_core.js); the Process engineering tab recomputes the design column from the live model and shows the checks.